

AKRAM FATAYER, Ph.D.

AI-Driven Neural Signal Processing · Human-Machine Interfaces · Translational Biomedical Engineering

From noisy physiological signals to validated, real-time clinical decisions

Research overview prepared for

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One core capability, three applications

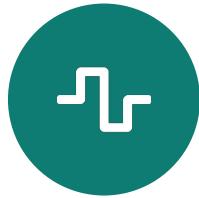
Across academia, industry, and clinical research, my work has solved a single recurring problem:

Extract reliable, real-time decisions from noisy, complex physiological signals — **and carry them through the full pipeline: algorithm → validated device → in-vivo evidence.**



DECODE

Decode motor intent from sEMG using deep learning (Ph.D., HIT)



MONITOR

Real-time neural monitoring in a Class III device (Double Medical)



MODULATE

Selective neuromodulation for therapeutics (research direction)

Decoding movement intent from messy biological signals

The problem

Surface EMG is noisy, non-stationary, and mislabeled — making robotic/prosthetic control unreliable in the real world.

My contribution

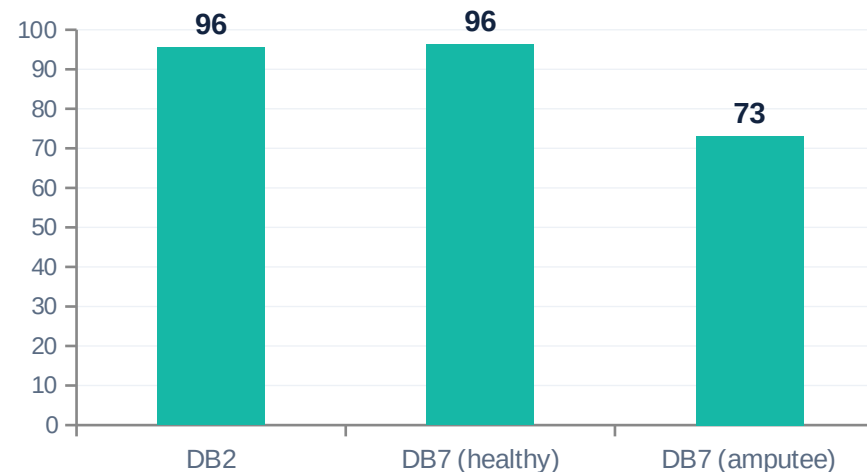
sEMG-Map — fuses raw sEMG with multi-resolution wavelet decomposition into a rich image the CNN can exploit.

ALR-CNN — a data-centric method that refines mislabeled samples while training, directly attacking label noise.

Published — IEEE J. Biomedical & Health Informatics (Q1)

Fatayer, Gao & Fu (2022) · 70+ citations · validated on 73 subjects, incl. 13 amputees

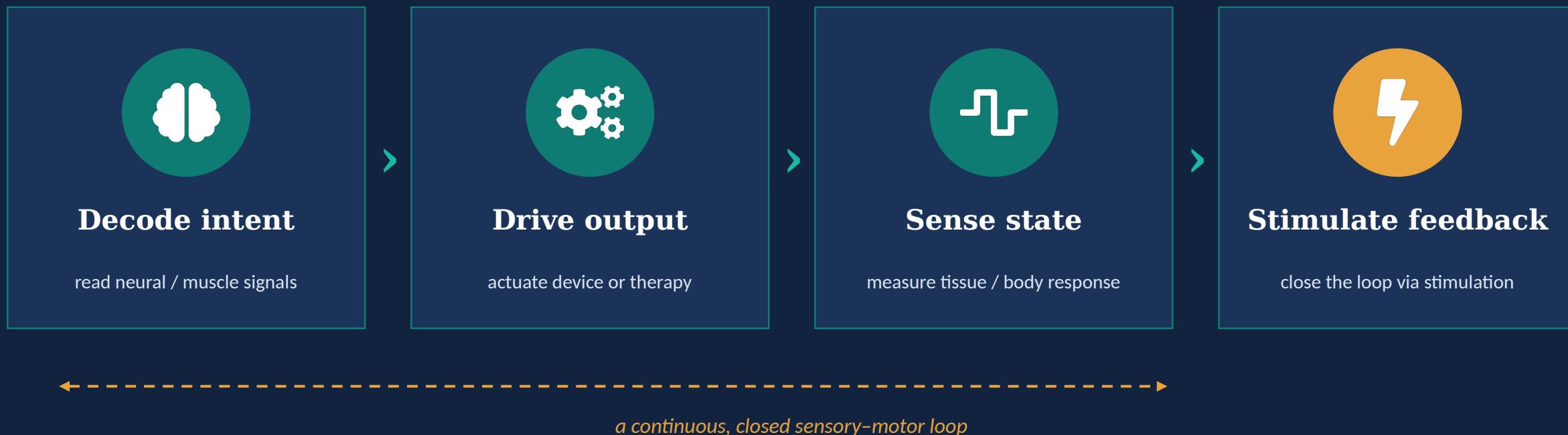
Accuracy across NinaPro benchmarks



+90% agreement with expert-labeled movement boundaries (ALR-CNN), a ~25% gain over baseline.

THE GUIDING IDEA

Decoding is only half the loop



My PhD solved the decoding side. The natural next step — and the thread connecting all my work — is **closing the loop with stimulation**: feedback, modulation, and therapy.

Taking algorithms all the way to the operating room

Real-time signal processing for a Class III intraoperative neuromonitoring (IONM) system — the full translational pipeline, under clinical and regulatory constraints.



1

Model & simulate

MATLAB/Simulink models of nerve & muscle response; synthetic signals with pink/white/powerline noise to stress-test robustness.



2

Denoise with AI

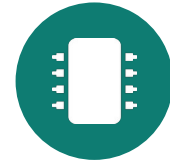
LSTM & RNN models to remove artifacts from EMG, MEP & SSEP recordings, improving device accuracy.



3

Validate in vivo

Hands-on porcine studies: nerve stimulation, electrophysiological recording, validation under surgical conditions.



4

Deploy to device

MATLAB-autogenerated C/C++ integrated into hardware; worked alongside surgeons & engineers.

Why it matters here: this is exactly the bench-to-clinic translational rigor — modeling, validation, regulatory-grade evidence — that precision medicine and clinical trials demand.

Neuromodulation as targeted therapy

Through an extended engagement with a neuroengineering group at Case Western Reserve University, I studied **photobiomodulation (PBM)** — using light to selectively modulate nerve activity. I recognized the experimental paradigm immediately: it mirrors my own preclinical stimulation work.



Selective pain modulation

PBM selectively inhibits small-diameter pain fibers (C, A δ) while sparing motor function — reducing nociceptive hypersensitivity by 56–70% in rodent models, without the side effects of opioids.



Neuroprotection

Near-infrared light (670 nm) delivered to the midbrain protected dopaminergic neurons and reduced clinical signs in a primate (MPTP) model of Parkinson's disease — a template for translation to clinical trials.

My fit: the design — induce a disease model, apply a stimulation protocol, measure electrophysiological & behavioral response — is the same paradigm I executed in porcine IONM studies. The stimulus changes; the rigor transfers.

THE CONVERGENCE

Three threads, one researcher



Decode

Deep learning on noisy biosignals — published, cited, benchmarked.



Monitor

Clinical-grade real-time neural monitoring in a Class III device.



Modulate

Targeted neuromodulation for pain & neurodegeneration therapy.

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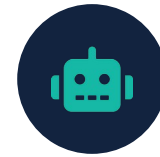
**An engineer who builds
the entire chain —
from raw signal to
validated therapy.**

Where my engine meets your research



Neuromuscular disease

My sEMG / motor-unit expertise maps directly onto motor-neuron disorders such as Spinal Muscular Atrophy — signal-based biomarkers and objective monitoring of disease & treatment response.



AI on noisy clinical data

Genomic, biomarker, and clinical datasets are high-dimensional and noisy. My noisy-label and signal-denoising methods translate naturally to robust clinical ML.



Trial-grade validation

I have taken algorithms through preclinical (in-vivo) validation to a regulated Class III device — the evidence pipeline that clinical translation requires.



Objective outcome measures

Real-time, quantitative physiological readouts can serve as objective endpoints for therapy and trial monitoring.

Let's explore where I fit

I bring a rare combination: **AI and signal-processing depth, hands-on preclinical experience, and full translational pipeline fluency** — applicable across neuromuscular disease, neuromodulation therapy, and AI-driven precision medicine.

Deep learning on noisy biosignals

Class III device & in-vivo validation

Neuromodulation for therapeutics

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